Scaling Effects of Neuronal Population Size on ECoG-Based Face and House Decoding

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Basis paper: *Spontaneous Decoding of the Timing and Content of Human Object Perception from Cortical Surface Recordings Reveals Complementary Information in the Event-Related Potential and Broadband Spectral Change*

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Abstract:

Previous studies (Miller et. al, 2016) have shown that it is possible to predict the type of stimuli (face, house, or other) using the broadband activity from face-selective and house-selective cells with up to 96% accuracy (of shown visual stimuli) with a less than 5% false positive rate and a ~20ms error in timing. This study shows how the responses of a neuronal population, measured through electrocorticography (ECoG), can be used to predict and decode the perceptual state of a patient with relatively high accuracy. The question we are seeking to answer is how decreasing the size of the neuronal population (resolution) affects the accuracy of the predictions, and at what rate the decline occurs. We aim to do this by first classifying which electrodes are "face-selective" and which electrodes are "house-selective." Then, we will reduce the size of the neuronal population by "lesioning" channels and graphing the resulting output. As a result, we can determine if rates of false positivity and errors in timing will increase as neuronal population decreases, even if the responses from the cells are not changing. This analysis examines whether the drop in accuracy scales proportionally with the rise in false positives, and whether that decline follows a linear or exponential trend. Using the ECoG dataset of epilepsy patients viewing images of faces and houses, we will first seek to reproduce the results seen in the previously mentioned study, using techniques such as power spectral analysis, and then map how the decline in resolution of the neuronal population corresponds to a decrease in accuracy of prediction. This dataset comes from seven patients with ECoG implants, over two sessions, sampled at a rate of 1000Hz. In the first task, subjects in a clinical setting were passively shown faces and houses. In the second task, the same subjects were shown face and house images with noise added and detection of faces was done by a keypress. Two of the subjects didn't have keypresses. This analysis could also reveal further insight into the lower bound of the necessary resolution of a neuron population needed to replicate the response seen in this dataset in a computational model.

References

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